

Module-3

Probability

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Outline

- 1 Introduction to Probability
- 2 Discrete Probability Models
- 3 Continuous Probability Models
- 4 Conditional Probability
- 5 Independence
- 6 Law of Total Probability and Bayes' Rule



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What is Probability?

Intuition

Probability measures the **likelihood of uncertain events**.

- It connects mathematics to real life: weather, gambling, stock market, reliability of machines.
- Always between 0 (impossible) and 1 (certain).
- Think of it as **long-run frequency** of an event if experiment is repeated many times.



Random Experiments

Definition

A **random experiment** is a process with an uncertain outcome. Each result is an **outcome**, and the set of all possible outcomes is the **sample space**.

Example

Toss a coin 3 times:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Outcome example: *HTT*.



Examples of Random Experiments

- Toss a coin: $S = \{H, T\}$.
- Roll a die: $S = \{1, 2, 3, 4, 5, 6\}$.
- Number of students absent in class: $S = \{0, 1, 2, \dots\}$.
- Rainfall in a day: $S = [0, \infty)$.



Definition

An **event** is a subset of the sample space. It represents a condition we are interested in.

Example

Roll a die. Event $E = \{\text{even number}\} = \{2,4,6\}$.



Event Types

- **Simple event:** consists of a single outcome, e.g., die roll = 5.
- **Compound event:** multiple outcomes, e.g., die roll even.
- **Impossible event:** empty set, probability = 0.
- **Certain event:** whole sample space, probability = 1.



Union and Intersection of Events

- **Union** ($A \cup B$): at least one of A or B occurs.
- **Intersection** ($A \cap B$): both A and B occur.
- **Complement** (A^c): event A does not occur.



Kolmogorov's Axioms of Probability

- ❶ For any event A , $\mathbb{P}(A) \geq 0$.
- ❷ $\mathbb{P}(S) = 1$.
- ❸ If $A_1, A_2, \dots$ are mutually exclusive (disjoint), then

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i) .$$



Derived Properties

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- $\mathbb{P}(\emptyset) = 0$.
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.



Takeaway

Probability theory = formalized rules for reasoning under uncertainty. Axioms guarantee consistency, like grammar rules in language.



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Discrete Sample Space

Definition

A sample space is **discrete** if outcomes can be counted.

$$\mathbb{P}(A) = \sum_{s \in A} \mathbb{P}(s)$$

Example

Roll a die: $S = \{1, 2, 3, 4, 5, 6\}$. Each outcome probability = $1/6$.



Finite Equally Likely Case

Formula

If sample space has N equally likely outcomes:

$$\mathbb{P}(A) = \frac{|A|}{N}.$$

Example

Die roll prime event $E = \{2, 3, 5\}$. $\mathbb{P}(E) = \frac{3}{6} = \frac{1}{3}$.



Example: Gambling Game (Problem)

Problem

In a game: win $k - 2$ rupees with prob $1/2^k$.

- Prob of winning 1–3 rupees?
- Prob of winning more than 2 rupees?



Example: Gambling Game (Solution)

Solution

Event $A = \{1, 2, 3\}$.

$$\mathbb{P}(A) = \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = \frac{7}{32} \approx 0.219$$

Event $B = \{3, 4, 5, \dots\}$.

$$\mathbb{P}(B) = \frac{1}{32} + \frac{1}{64} + \frac{1}{128} + \dots = \frac{1}{16} = 0.0625$$



Concept Highlight

Key Idea

Discrete probabilities often involve **counting** or **series sums**. Infinite but countable sample spaces are common in gambling or waiting-time models.



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Continuous Sample Space

Definition

When outcomes are real numbers in an interval, sample space is **continuous**.

- Probability of single point = 0.
- Probabilities are computed as **areas under a curve (integrals)**.



Uniform Example (Problem)

Friend arrives uniformly between 1–2 pm.

- 1 What is $\mathbb{P}(T = 1.5)$?
- 2 What is $\mathbb{P}(T \in [1, 1.5))$?



Uniform Example (Solution)

Solution

$$\mathbb{P}(T = 1.5) = 0. \quad \mathbb{P}(T \in [1, 1.5)) = 0.5.$$

$$\text{General case: } \mathbb{P}([a, b]) = b - a.$$



Concept Highlight

Key Idea

Continuous probability = limit of discrete. Probability of exact number = 0, but intervals matter.



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Conditional Probability: Definition

Formula

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \mathbb{P}(B) > 0$$

Intuition

Think of **shrinking sample space** to cases where B happens. Now measure chance of A inside that smaller world.



Real-Life Intuition

- Probability of rain on any day = 0.2.
- If sky is cloudy, probability of rain rises to 0.7.
- Cloudy acts as extra information $\rightarrow$ updates probability.



Dice Example (Problem)

Roll 2 dice: (X_1, X_2) . Given $X_1 + X_2 = 7$, find probability $X_1 = 4$ or $X_2 = 4$.



Dice Example (Solution)

Solution

Possible sums=7: (6,1),(5,2),(4,3),(3,4),(2,5),(1,6). Favorable: (4,3),(3,4).

$$\mathbb{P}(A|B) = \frac{2/36}{6/36} = \frac{1}{3}$$



Children Example

Sample space = $\{GG, GB, BG, BB\}$.

Q1: Both girls given first is girl. **Q2:** Both girls given at least one girl.



Children Example (Solution)

- Q1: $\mathbb{P}(GG|firstG) = \frac{1}{2}$.
- Q2: $\mathbb{P}(GG|atleastoneG) = \frac{1}{3}$.



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Independence of Events

Definition

Events A, B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B) .$$

- Example: Coin toss vs tomorrow's weather.



Independence Example

Pick number from 1–10. $A = \{N < 7\}$, $B = \{N \text{ even}\}$.

Find $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$



Independence Example (Cont'd)

Pick number from 1–10. $A = \{N < 7\}$, $B = \{N \text{ even}\}$.

Solution

$\mathbb{P}(A) = 0.6$, $\mathbb{P}(B) = 0.5$, $\mathbb{P}(A \cap B) = 0.3$. (Since $0.3 = 0.6 \cdot 0.5$, they are independent.)



Disjoint vs Independent

Important

- **Disjoint:** cannot happen together.
- **Independent:** happening of one doesn't affect probability of other.



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Question

How do we calculate probability of an event A if it can happen in many different ways?

- Example: Probability a student passes exam depends on which coaching class they joined.
- Break down event A into disjoint cases $B_1, B_2, \dots$
- This leads to the **Law of Total Probability**.



Law of Total Probability (Derivation)

Suppose $B_1, B_2, \dots, B_n$ partition the sample space S :

$$S = B_1 \cup B_2 \cup \dots \cup B_n, \quad B_i \cap B_j = \emptyset.$$

Then for any event A :

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n).$$

By additivity of probability:

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i).$$

Using conditional probability:

$$\mathbb{P}(A \cap B_i) = \mathbb{P}(A|B_i) \mathbb{P}(B_i).$$

Hence,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i) \mathbb{P}(B_i).$$



Law of Total Probability (Intuition)

Key Idea

Probability of A can be computed by:

- 1 Splitting the world into cases $B_1, B_2, \dots$,
- 2 Computing probability of A in each case,
- 3 Weighting by chance of each case.



Example: Die Roll

Problem

A fair die is rolled. Let A = event “number is even”. Partition by whether die is ≤ 3 or > 3 : $B_1 = \{1, 2, 3\}$, $B_2 = \{4, 5, 6\}$. Use total probability to find $\mathbb{P}(A)$.



Example: Die Roll (Solution)

Solution

$$\mathbb{P}(A) = \mathbb{P}(A|B_1) \mathbb{P}(B_1) + \mathbb{P}(A|B_2) \mathbb{P}(B_2) .$$

$$\mathbb{P}(B_1) = 0.5, \mathbb{P}(B_2) = 0.5.$$

In B_1 : $A = \{2\}$, so $\mathbb{P}(A|B_1) = 1/3$.

In B_2 : $A = \{4, 6\}$, so $\mathbb{P}(A|B_2) = 2/3$.

Therefore:

$$\mathbb{P}(A) = (1/3)(0.5) + (2/3)(0.5) = 0.5$$

which matches direct calculation.



Bayes' Rule (Derivation)

From definition of conditional probability:

$$\mathbb{P}(B_j|A) = \frac{\mathbb{P}(A \cap B_j)}{\mathbb{P}(A)}.$$

But $\mathbb{P}(A \cap B_j) = \mathbb{P}(A|B_j) \mathbb{P}(B_j)$.

Also, by total probability:

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A|B_i) \mathbb{P}(B_i) .$$

Hence,

$$\mathbb{P}(B_j|A) = \frac{\mathbb{P}(A|B_j) \mathbb{P}(B_j)}{\sum_i \mathbb{P}(A|B_i) \mathbb{P}(B_i)} .$$



Bayes' Rule (Intuition)

Key Idea

Bayes' rule lets us **reverse conditioning**: from $\mathbb{P}(A|B)$ (likelihood) to $\mathbb{P}(B|A)$ (posterior).

Think:

- Prior belief: $\mathbb{P}(B_j)$
- Evidence: $\mathbb{P}(A|B_j)$
- Updated belief: $\mathbb{P}(B_j|A)$



Example: Marble Problem

3 bags equally likely.

- Bag1: 75 red, 25 blue
- Bag2: 60 red, 40 blue
- Bag3: 45 red, 55 blue

One bag chosen at random, then a marble drawn. Given marble is red, find $\mathbb{P}(\text{Bag1} | R)$.



Marble Example (Solution)

Solution

$$\mathbb{P}(R|B_1) = 0.75, \mathbb{P}(R|B_2) = 0.60, \mathbb{P}(R|B_3) = 0.45$$

$$\mathbb{P}(R) = (0.75 + 0.60 + 0.45)/3 = 0.60$$

So,

$$\mathbb{P}(B_1|R) = \frac{0.75 \cdot (1/3)}{0.60} = \frac{5}{12}.$$



Example: Medical Test (Problem)

A disease affects 1 in 1000 people. A test has:

- Sensitivity = 99% (positive if diseased),
- Specificity = 95% (negative if healthy).

If a person tests positive, what is probability they really have disease?



Medical Test (Solution)

Solution

Let D =disease, T =positive test.

$$\mathbb{P}(D) = 0.001, \mathbb{P}(D^c) = 0.999$$

$$\mathbb{P}(T|D) = 0.99, \mathbb{P}(T|D^c) = 0.05$$

By Bayes:

$$\begin{aligned}\mathbb{P}(D|T) &= \frac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.05 \cdot 0.999} \\ &= \frac{0.00099}{0.00099 + 0.04995} \approx 0.019\end{aligned}$$

So even with a positive test, chance is only 1.9%.



Takeaway

- Law of Total Probability: split by cases, then add.
- Bayes' Rule: update beliefs with evidence.
- In medical tests, even accurate tests can have many false positives if disease is rare!

